「Quotient Factorization.

「Thm. Let $\varphi: G \to H$ be a group Homomorphism, and $N \triangleleft G$ be a normal. Then, $N \subseteq \ker\varphi$ if and only if there exists a unique Homomorphism

$$\overline{\varphi}: G/N \to H \text{ Such that } \varphi = \overline{\varphi} \circ \pi$$

where the $\pi: G \to G/N: g \mapsto gN$ is the Canonical Prodection.

Proof. Suppose that $N \subseteq \ker\varphi$, Define

$$\overline{\varphi}: G/N \to H: gN \mapsto \varphi(g).$$

It is well-defined, because

$$g_1N = g_2N \iff g_1^{-1}g_2 \in N \Rightarrow g_1^{-1}g_2 \in \ker\varphi \iff \varphi(g_1^{-1}g_2) = e \iff \varphi(g_1) = \varphi(g_2)$$

And, Since φ preserves the operation, so $\overline{\varphi}$ is Homomorphism.

And, given $g \in G$, $(\overline{\varphi} \circ \pi)(g) = \overline{\varphi}(gN) = \varphi(g)$, so $\overline{\varphi} \circ \pi = \varphi$.

To show the uniqueness, let $\psi: G/N \to H$ be a map satisfying $\psi \circ \pi = \varphi$.

Given $gN \in G/N$, $\psi(gN) = \psi(\pi(g)) = (\psi \circ \pi)(g) = \varphi(g) = \overline{\varphi}(gN)$,

therefore $\psi = \overline{\varphi}$.